CHANDU PRASAD BHAIRAPU



AI Engineer

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ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in building data and automation systems end to end, from system design and Python backends with APIs to LLM-powered agents and cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector at an early-stage start-up, backed by a strong foundation in Python development and AI-assisted software engineering.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer with full ownership of the architecture of an agentic knowledge-extraction system end to end, from data ingestion and chunking to prompt engineering and indexing (agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Made a product-driven technical decision for a second, high-volume source stream: built a deterministic, code-based extraction pipeline instead of routing it through LLMs, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed the logic of a multi-agent extraction system with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents and set quality standards by measuring it against an SAP-system-derived ground truth, driving precision/recall from 33%/45% to 93%/90%.
- Designed and implemented a Model Context Protocol (MCP) query service as the API to the knowledge graph, with millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts validate and approve knowledge-graph changes.
- Set up an AI-assisted development workflow (Claude Code with custom skills, agentic coding) grounded in clean code principles.
- **Tech stack: Python, FastAPI, MCP, LangGraph, LiteLLM, Docker, Kubernetes, Azure, Postgres, MongoDB, Neo4j**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked directly with business stakeholders to understand their decision-making workflows, advised on where agentic AI could support them, and translated their requirements into a concrete solution approach.
- Designed and deployed an agentic AI application with LangGraph whose routing logic sends each request to vector database queries, general inference, or summarization.
- Built the Python FastAPI backend (API testing in Postman) serving a React frontend, containerized the stack with Docker, and deployed it to AWS with CDK and GitLab CI/CD.
- **Tech stack: Python, FastAPI, LangGraph, LangChain, React, Docker, AWS (CDK, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of answer quality and retrieval for the PrismaChat RAG system with RAGAS and PytrecEval against chemist-provided ground truth, on AWS infrastructure (Lambda, SQS, DynamoDB) built with CDK.
- Designed RAG data pipelines (chunking strategies, vector databases, indexing, fine-tuning models for better retrieval) and benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, AWS (Lambda, SQS, DynamoDB, CDK), RAGAS, PytrecEval, LangChain, PyTorch, Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, engineering lag features and rolling statistical windows from sensor data.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Client Occasion Engine | Python | FastAPI | Postgres | LLM API

- API-first backend that scans a synthetic insurance-broker client portfolio for service occasions (premium increase, underinsurance, coverage gap), prioritises them, and routes each client to the next action and channel through a configurable trigger and rule layer; an LLM drafts the personalised service message.
- Integrations: an adapter syncing client data from a mock broker management system over REST; an event log of every trigger and outcome drives monitoring and per-rule learning curves; tests run in GitHub Actions CI.

Project: Spec-Driven Coding Agent Harness | Claude Code | FastMCP | GitHub Actions

- Harness that turns a plain-language feature request into a testable specification (acceptance criteria written as pytest tests) and lets Claude Code implement it under a rule set: project conventions, custom skills, and hooks that block edits to protected files and secrets; an MCP server built with FastMCP gives the agent the spec and task context.
- A CI pipeline gates every agent-written change with automated tests, security and dependency scans, and a benchmark evaluation suite.

TECHNICAL SKILLSET

- **Backend & APIs:** Python, FastAPI, REST APIs, Python microservices, MCP servers, SQL, Postman
- **Data & Automation:** data pipelines (ingestion, chunking, indexing), workflow orchestration and routing (LangGraph), Postgres, MongoDB, Neo4j, DynamoDB, Pandas
- **Cloud & Delivery:** Docker, Kubernetes, Azure, AWS (CDK, Lambda, SQS, EC2, S3), GitLab CI/CD, Git, Agile
- **AI-Assisted Development:** Claude Code (custom skills, agentic coding workflows, MCP integration)
- **AI & LLM Systems:** LLM APIs (LiteLLM), LangChain, multi-agent systems, RAG, GraphRAG, prompt engineering, evaluation (RAGAS, PytrecEval), Hugging Face Transformers, fine-tuning (QLoRA)
- **Frontend & Machine Learning:** React, Streamlit, Scikit-Learn, XGBoost, PyTorch, TensorFlow

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)